

Article

Does Performance Feedback Drive Greenwashing and Brownwashing? Evidence from China's Capital Market

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Abstract

Against the policy backdrop of high-quality development and the “Dual Carbon” goals, corporate environmental responsibility and green governance have emerged as core drivers of corporate value creation and resource allocation in capital markets. However, in practice, corporate environmental disclosure has increasingly degenerated into an impression management tool. Using a sample of China's A-share listed companies from 2011 to 2024, this paper combines text analysis of annual reports with green patent data to systematically examine the impact of performance feedback on corporate strategic environmental decoupling, drawing upon the behavioral theory of the firm and legitimacy theory. The findings are as follows: First, negative performance feedback significantly increases corporate greenwashing propensity, whereas positive performance feedback significantly strengthens corporate brownwashing behavior. Second, government regulation amplifies the costs of falsifying environmental information, significantly suppressing the positive impact of negative performance feedback on greenwashing, but exacerbating the positive impact of positive performance feedback on brownwashing. Conversely, media attention amplifies the benefits of corporate green performances, significantly strengthening the catalytic effect of negative performance feedback on greenwashing, while effectively suppressing the positive impact of positive performance feedback on brownwashing. Third, heterogeneity analysis reveals that the impact of performance feedback on corporate strategic decoupling in environmental disclosure is more pronounced among non-state-owned enterprises, firms facing high industry competitive pressure, and those in heavily polluting industries. By integrating greenwashing and brownwashing into a unified analytical framework, this study expands the research boundaries of corporate environmental disclosure and strategic behaviors. Furthermore, it deepens the application contexts of the behavioral theory of the firm within non-financial disclosure, deconstructs the myth of homogeneous governance effects under legitimacy pressure, and provides vital implications for investors, policymakers, and fund managers.



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1. Introduction

As the Chinese economy transitions toward a stage of high-quality development, promoting the green and low-carbon transformation of economic and social development has emerged as a critical component of advancing Chinese modernization. The

15th Five-Year Plan explicitly proposes to “accelerate the comprehensive green transformation of economic and social development and build a Beautiful China.” Against this backdrop, sustainable development concepts such as ESG (Environmental, Social, and Governance) have evolved into a core criterion determining corporate financing costs and market valuations. As of the third quarter of 2024, the scale of China’s ESG investment market has reached an impressive 40.31 trillion RMB. (By the End of the Third Quarter, the Scale of China’s Responsible Investment Market Exceeded 40 Trillion RMB, China Securities Journal Online, <https://www.cnstock.com/commonDetail/324789>, accessed on 24 November 2025) However, with the influx of capital, environmental information disclosure has gradually deviated from its original intent of responsibility fulfillment, degenerating into a tool for corporate “impression management.” EY’s 2024 Global Institutional Investor Survey reveals that 85% of investors believe misleading statements, such as greenwashing, have become more severe compared to five years ago (2024 Global Institutional Investor Survey, EY, https://www.ey.com/en_gl/insights/climate-change-sustainability-services/institutional-investor-survey) accessed on 24 November 2025. Currently, academia and regulatory bodies predominantly focus on such “overstated” greenwashing behaviors, yet they overlook another equally prevalent but more insidious phenomenon in the capital market: some firms have made substantial green technological innovations and environmental protection investments, but deliberately maintain a low profile and remain silent during disclosure—a practice known as brownwashing. The coexistence of such “overstating” and “understating” suggests that corporate information manipulation on environmental issues is not merely a unidirectional ethical lapse, but rather a two-way strategic environmental decoupling encompassing both “greenwashing” and “brownwashing.” What underlying logic drives firms to adopt such diametrically opposed strategies of “decoupling words from deeds” in environmental disclosure? What role does external legitimacy pressure play in this process? Systematically addressing these questions not only helps mitigate the risks of strategic disclosure in the capital market but also holds profound practical significance for refining ESG regulatory policies.

The academic community has devoted considerable attention to environmental issues. Scholars have identified the decoupling of words and deeds in corporate environmental disclosure and practices, conceptualizing these phenomena as strategic disclosure behaviors such as greenwashing and brownwashing [1]. Existing research has explored the antecedents of environmental decoupling from multiple perspectives, which can primarily be categorized into two streams: the first focuses on internal corporate motives, such as managerial myopia [2], top management team (TMT) stability [3], and the pursuit of profit maximization [4]; the second emphasizes external institutional deficiencies, such as the lack of unified environmental disclosure standards [5], insufficient regulatory penalties and regulatory pressures [6]. Despite the extensive literature on the drivers of corporate environmental decoupling, two critical theoretical blind spots remain. First, prior studies tend to treat ESG disclosure as a static moral choice, neglecting its fundamental nature as a tool for impression management [7]. According to the behavioral theory of the firm, corporate strategic choices are profoundly shaped by their financial performance aspiration discrepancies [8]. However, scant research has explored how a firm’s internal survival conditions and cost–benefit expectations drive its propensity for strategic environmental decoupling when navigating favorable circumstances (positive performance feedback) versus adversity (negative performance feedback) [9]. Second, when examining external governance mechanisms, the existing literature often makes the sweeping assumption that legitimacy pressures (e.g., from the government and the media) will inevitably deter corporate misconduct [10]. However, from a cost–benefit perspective, all information games in human society ultimately depend on the margin between the costs and benefits

of falsification. External pressures do not invariably act as monitors; depending on their specific power attributes, they may alter the costs or benefits of manipulation, thereby influencing corporate decision-making.

Building on this, drawing upon the behavioral theory of the firm and legitimacy theory, this paper constructs an analytical framework of “performance feedback—legitimacy pressure—corporate environmental decoupling.” This paper argues that strategic environmental decoupling serves as a reflection of a firm’s survival conditions and an impression management strategy under varying performance feedback. When facing negative performance feedback, driven by intense survival anxiety, management tends to adopt a greenwashing strategy, attempting to acquire legitimacy resources at minimal public relations costs to mask financial failures. Conversely, when receiving positive performance feedback, to evade shareholder criticism regarding agency costs or the external pressure of “whipping the fast ox” (imposing heavier burdens on high performers), management is inclined to adopt an “understating” brownwashing strategy for strategic invisibility. Furthermore, from the perspective of the costs and benefits of misrepresentation, this paper profoundly reveals how legitimacy pressures from the government and the media influence corporate environmental decoupling. On the one hand, government regulation, endowed with auditing and punitive powers, significantly elevates the costs of corporate misrepresentation. Consequently, stringent government regulation not only curbs the greenwashing propensity of underperforming firms but also further amplifies the motivation of high-performing firms to avert political risks, thereby exacerbating their brownwashing behaviors. On the other hand, while the media can heighten public attention toward a firm, it lacks verification authority; thus, it substantially increases the benefits of misrepresentation without materially raising its costs. As a result, media attention can trigger the greenwashing impulses of underperforming firms, while simultaneously dismantling the strategic invisibility of high-performing firms, thereby suppressing their brownwashing propensity.

This paper selects the Chinese capital market as the research context primarily based on its uniqueness and typicality in environmental governance and the institutional environment. First, China is at a critical juncture of transitioning its economy toward high-quality development, and the introduction of the “Dual Carbon” goals has subjected enterprises to unprecedented environmental legitimacy pressure. Second, the Chinese capital market features unique government-business relations and ownership structures, resulting in significant heterogeneity between state-owned and non-state-owned enterprises regarding their cost–benefit expectations when acquiring environmental legitimacy resources. Third, China’s green information disclosure system is in a crucial transition period from voluntary to mandatory, making issues of information asymmetry and strategic disclosure particularly prominent.

The findings of this study contribute to the literature in three primary ways. First, this study expands the research boundaries of environmental information disclosure by constructing an analytical framework of strategic decoupling in environmental disclosure that encompasses both greenwashing and brownwashing. Breaking away from the singular focus on greenwashing, this paper integrates the capital market context to profoundly reveal how brownwashing, as a concealed impression management tool, exacerbates information asymmetry and triggers value penalties for green assets. This provides a novel theoretical perspective for comprehensively understanding the phenomenon of corporate environmental decoupling. Second, this study deepens the application contexts of the behavioral theory of the firm by revealing, for the first time, the asymmetric logic of environmental information manipulation. Traditional research on the behavioral theory of the firm has primarily confined the analytical framework of performance aspiration discrepancies to the allocation

of physical resources, such as R&D and mergers and acquisitions (M&A), paying less attention to strategic behaviors in the realm of non-financial information. This paper extends this framework to the domain of environmental impression management, precisely delineating the distinctly different survival conditions and cost–benefit expectations of firms under positive and negative performance feedback. Third, this study deconstructs the myth of homogeneous governance effects of legitimacy pressure within neo-institutional theory. Existing literature largely assumes that external monitoring (e.g., by the government and the media) inevitably exerts a unidirectional, positive governance effect on corporate compliance behaviors. From the perspective of the actuarial cost–benefit analysis of information manipulation, this paper clarifies the fundamental differences between these two core types of legitimacy pressure. This study posits that government regulation acts as an amplifier for the costs of falsifying environmental information, whereas media attention serves as an amplifier for the benefits of “green acting.” This paradigm shift elegantly explains why external monitoring may not only fail in specific contexts but even induce more severe greenwashing performances or exacerbate strategic invisibility, providing breakthrough theoretical insights for understanding government-business and media-business games in complex institutional environments. In terms of practical implications, for regulatory authorities, it is necessary to not only prevent the exaggerated greenwashing behaviors of distressed firms but also pay attention to the brownwashing issues of high-performing firms, thereby avoiding the distortion of environmental information disclosure quality. Simultaneously, regulatory tools should be applied differentially: utilizing government regulation to precisely target greenwashing behaviors, while employing guiding policies to encourage high-performing firms to reasonably disclose their substantive green innovation achievements. For the media, there is a need to enhance professional capabilities in reporting on environmental issues, shifting the focus from the quantity of corporate textual disclosure to the quality of substantive green practices, to avoid becoming an enabler of corporate greenwashing performances.

2. Literature Review and Research Hypotheses

2.1. Performance Feedback and Corporate Environmental Decoupling

With the widespread adoption of sustainable development concepts, the authenticity and effectiveness of corporate environmental information disclosure have increasingly become a focal point of attention for both academia and practice, among which strategic decoupling in environmental disclosure has emerged as a major area of controversy. Based on the divergence between public disclosure and actual implementation, corporate strategic decoupling in environmental disclosure can be categorized into the following four types [1]: (1) exaggerating green performance while concealing negative information; (2) failing to fulfill committed environmental targets [11]; (3) deliberately downplaying or evading information regarding environmental commitments and practices; and (4) systematically underreporting environmental achievements, demonstrating strategic modesty [12]. The first two types of behavior are typically classified as greenwashing, whereas the latter two are regarded as brownwashing.

Existing literature predominantly views ESG washing as a corporate moral issue or a static form of impression management, neglecting the fact that firms do not make decisions in a vacuum; rather, their ESG information disclosure strategies are profoundly influenced by their current financial conditions. The behavioral theory of the firm posits that corporate strategic choices are highly contingent upon the comparison between actual performance and performance aspirations. When actual performance falls below aspirations (negative performance feedback), firms perceive a survival crisis and trigger problemistic search. In such scenarios, they prioritize low-cost, quick-yielding short-term strategies to rapidly

repair the performance shortfall and legitimacy crisis, leading to a significant increase in their risk-taking propensity. Conversely, when actual performance exceeds aspirations (positive performance feedback), firms possess abundant slack resources and the pressure for survival dissipates. Consequently, their core decision-making objective shifts toward consolidating long-term competitive advantages, making them more inclined to adjust long-term resource allocation strategies based on factors such as risk aversion and the protection of technological barriers.

The fundamental motivation for firms to actively undertake social responsibility stems from their need for legitimacy. Legitimacy theory posits that a firm's survival and development depend on stakeholder recognition and congruence with social norms, making legitimacy a core scarce resource. Consequently, corporate operational activities and information disclosure behaviors are fundamentally means to acquire, maintain, or repair this legitimacy. When a firm's operational activities deviate from stakeholder expectations or social norms, it faces legitimacy pressure. The intensity of this pressure directly determines the urgency and direction of the firm's strategic adjustments. Legitimacy pressure primarily originates from two major channels. The first is coercive legitimacy, which stems from government regulation and the mandatory requirements of laws and regulations, characterized by its compulsory and punitive nature. The second is normative legitimacy, which arises from the attention and expectations of stakeholders such as the media, the public, and investors, characterized by its market-driven and public opinion-oriented nature. Within the research context of this paper, government regulation serves as the core source of coercive legitimacy, while media supervision acts as the core source of normative legitimacy.

Based on the behavioral theory of the firm and prospect theory, this paper argues that environmental decoupling serves as a strategic tool for firms to engage in legitimacy hedging and expectation management under varying performance feedback. Negative performance feedback indicates that a firm's actual performance falls below its endogenous performance aspirations, plunging the firm into dual financial and legitimacy pressures. From the perspective of financial pressure, the firm must concentrate its core resources to repair the performance shortfall; consequently, resources are channeled toward core businesses that can directly improve profitability, making it difficult to sustain high-cost, long-cycle substantive environmental investments. From the perspective of legitimacy pressure, the performance shortfall triggers negative expectations among core stakeholders, such as investors and creditors, thereby exacerbating the firm's legitimacy crisis and necessitating a rapid restoration of its market image. At this juncture, the firm initiates problemistic search, prioritizing low-cost, quick-yielding short-term strategies. Greenwashing requires only selective disclosure and the exaggeration of green commitments to rapidly project a positive green image to the market and alleviate legitimacy pressure, all without incurring the high costs of substantive environmental investments and green R&D. Therefore, it constitutes a rational choice for firms operating under negative performance feedback.

Positive performance feedback signifies that a firm's actual performance exceeds its aspirations. In this scenario, the firm possesses abundant slack resources, financial pressure dissipates, and its core decision-making objective shifts toward consolidating long-term competitive advantages. The availability of ample slack resources enables the firm to undertake high-cost, long-cycle substantive green innovations and environmental investments, laying the foundation for "doing more." First, avoiding shareholder resistance to excess green investments and agency conflicts. Existing research indicates that avoiding shareholder resistance to green investments is a core financial motivation for firms to choose strategic silence). Substantive green innovations are often characterized

by large investments, long cycles, and high uncertainty [13]. For highly profitable firms receiving positive performance feedback, if management heavily touts their massive excess investments in the non-financial (environmental) domain in annual reports, it is highly likely to trigger concerns among external shareholders. Shareholders may perceive management as squandering free cash flows that rightfully belong to them to build a personal “green empire” (Empire building), thereby crowding out resources for other high-return projects. Second, guarding against “hypocrisy accusations” and the reputational backlash brought about by high expectations. Promoting green achievements can easily elevate the moral expectations of stakeholders. As Carlos and Lewis (2018) point out, when firms perceive a potential threat of “hypocrisy,” they often deliberately conceal their sustainable achievements [14]. For high-performing firms, they are already under the market’s spotlight and enjoy a relatively high initial reputation. If their overly high-profile environmental commitments fail to materialize in the future due to minor environmental accidents or technological iteration failures, these high expectations will instantly transform into severe “hypocrisy accusations” and “reputational backlash.” Therefore, under positive performance feedback, firms rationally opt for an “understating” brownwashing strategy. While making substantive ESG investments, they proactively reduce related disclosures to protect technological barriers and avert potential risks. Therefore, we propose the following hypotheses:

H1a. *Negative performance feedback significantly increases a firm’s greenwashing propensity.*

H1b. *Positive performance feedback significantly increases a firm’s brownwashing propensity.*

2.2. The Moderating Effect of Legitimacy Pressure

The governance effect of legitimacy pressure depends on how it alters a firm’s costs and benefits of misrepresentation. Government regulation reflects the local government’s prioritization of environmental issues and the stringency of its enforcement. Endowed with auditing and punitive powers, it significantly elevates the costs of corporate misrepresentation in environmental information disclosure [15]. On the one hand, for underperforming firms mired in negative performance feedback, although they possess a strong motivation to greenwash to acquire a green reputation, the looming regulatory pressure in a highly stringent government regulatory environment makes them realize that once greenwashing behaviors are detected and penalized, the firm will face massive fines or even suspension of operations for rectification. In such cases, the exorbitant costs of misrepresentation far outweigh the potential legitimacy benefits, thereby generating a potent deterrent effect. Consequently, government regulation effectively suppresses the greenwashing attempts of underperforming firms. On the other hand, for high-performing firms receiving positive performance feedback, the intensification of macro-level government regulatory pressure implies that local governments are in urgent need of identifying high-quality enterprises to shoulder regional environmental and social governance tasks. At this juncture, to avoid becoming targets of public scrutiny and victims of “whipping the fast ox” (i.e., penalizing high performers with extra burdens), the motivation of high-performing firms to deliberately conceal their substantive green achievements is further reinforced. Therefore, the more stringent the government regulation, the more intensely the strategic invisibility motives of high-performing firms—feigning poverty and maintaining a low profile—are stimulated. Accordingly, this paper proposes: Therefore, we propose the following hypotheses:

H2a. *Government regulation negatively moderates the positive relationship between negative performance feedback and corporate greenwashing.*

H2b. *Government regulation positively moderates the positive relationship between positive performance feedback and corporate brownwashing.*

Distinct from government regulation, media attention essentially acts as a spotlight. The media possesses formidable capabilities in agenda-setting and arousing public sentiment, yet it lacks the authority to verify firms' underlying data and impose penalties. Consequently, media attention substantially elevates the benefits of corporate misrepresentation in environmental information without materially increasing its costs. On the one hand, for underperforming firms mired in negative performance feedback, the media spotlight provides a perfect stage for performative actions. Under intense media scrutiny, public sensitivity to corporate environmental dynamics is exceptionally high. This maximizes the brand premium and sympathetic goodwill that firms can extract by issuing exaggerated public relations statements (greenwashing). To cater to the media's demand for short-term narratives, underperforming firms will engage in superficial window-dressing even more aggressively. Therefore, the media paradoxically induces more severe greenwashing. On the other hand, for high-performing firms receiving positive performance feedback, their initial intent is to adopt a brownwashing strategy for strategic invisibility. However, exceptionally high media attention completely shatters this cloak of invisibility. Under the spotlight, both the firm's financial condition and its social responsibility performance are subjected to public scrutiny. If a firm boasts excellent financial performance but deliberately conceals its green contributions, it may instead be construed by the public and the media as "rich but malevolent" or lacking in social responsibility. If a highly profitable firm remains silent on environmental issues before the media, it will immediately trigger public moral skepticism and a reputational crisis. To quell the pressure of public opinion, high-performing firms are compelled to abandon their invisibility strategy and publicly disclose their authentic ESG achievements. Accordingly, this paper proposes: Therefore, we propose the following hypotheses:

H3a. *Media attention positively moderates the positive relationship between negative performance feedback and corporate greenwashing.*

H3b. *Media attention negatively moderates the positive relationship between positive performance feedback and corporate brownwashing.*

3. Research Design

3.1. Variable Selection and Measurement

3.1.1. Dependent Variable

Existing research primarily employs two methods to measure corporate environmental decoupling: the first constructs a washing index based on the discrepancy between a firm's actual environmental performance and its external information disclosure [16]; the second relies on text analysis techniques to identify inconsistencies between a firm's environmental information disclosure content and its actual behaviors [17]. Because the core of strategic decoupling in environmental disclosure lies in the systematic deviation between disclosed information and authentic performance, measuring it solely from a single perspective—such as disclosure tone or rating scores—often fails to accurately capture this "decoupling of words and deeds." In contrast, the essence of corporate strategic decoupling in environmental disclosure is the strategic decoupling of a firm's "words" (disclosure)

and “deeds” (practice) regarding environmental and social responsibility issues. Drawing on the approaches of Zeng et al. (2024) and Chen et al. (2025), this paper independently measures corporate “words” and “deeds” to construct a continuous index of strategic decoupling in environmental disclosure [18,19]. As statutory disclosure documents strictly regulated by the China Securities Regulatory Commission (CSRC), annual reports represent the highest level of corporate strategic will. If a firm still exhibits significant “decoupling of words and deeds” in its rigorously audited annual reports, it provides compelling evidence that such decoupling behavior is not merely short-term packaging by the public relations department, but rather a deliberate strategic manipulation by management.

This paper employs text analysis to characterize firms’ willingness to disclose in the environmental domain. Regarding text sources, this study selects the “Management Discussion and Analysis (MD&A)” section in the annual reports of A-share companies listed on the Shanghai and Shenzhen Stock Exchanges in China. Compared to the full annual report, the MD&A section more centrally reflects the strategic intentions and future planning of the highest decision-making level, effectively filtering out the interference of mechanical texts such as notes to financial statements. Specifically, referring to Zeng et al. (2024) Chen et al. (2025) and the policy documents of the Ministry of Ecology and Environment and relevant classic literature, this paper constructs a systematic and exclusive “Green Environmental Dictionary” [18,19]. This dictionary comprehensively covers the following four core environmental issues: First, climate change and carbon reduction, including low-carbon, carbon neutrality, carbon peaking, carbon emissions, carbon footprint, greenhouse gases, climate change, etc.; Second, pollution prevention and control, including environmental protection, emission reduction, pollution treatment, pollution prevention, desulfurization, denitrification, dust removal, sewage treatment, exhaust gas, solid waste, etc.; Third, resources and clean energy, including energy conservation, consumption reduction, water saving, clean energy, renewable energy, green energy, circular economy, comprehensive utilization of resources, etc.; Fourth, green strategy and ecology, including green, green transformation, green manufacturing, green factories, green supply chains, ecological protection, ecological restoration, environmental management, sustainable development, etc. This study utilizes Python software 3.12.9 to calculate the frequency of relevant keywords in the MD&A section of annual reports to measure firms’ willingness to disclose environmental information. To effectively eliminate the mechanical confounding effect of document length and the overall length of the annual report, this paper adopts relative word frequency to measure the intensity of corporate environmental disclosure. Specifically, this paper divides the total frequency of extracted green keywords in the MD&A section by the total number of words in the annual report, yielding the firm’s “words” indicator (denoted as Talk). To objectively measure a firm’s substantive efforts in environmental protection, this paper discards easily manipulated soft metrics and selects green technological innovation as the proxy variable for “deeds.” The R&D and application of green patents require firms to invest substantial financial and R&D resources, and they must undergo rigorous examination by the China National Intellectual Property Administration (CNIPA). Thus, they represent the most core and hardest-to-fake green practices in a firm’s pursuit of green development. This paper takes the natural logarithm of the number of green patents independently applied for and ultimately granted to the firm in the current year plus one, yielding the firm’s “deeds” indicator (denoted as Action). Because “words” and “deeds” belong to entirely different scales, direct mathematical operations cannot be performed. To achieve comparability between the two and to eliminate the confounding effects of industry heterogeneity and time trends, this paper standardizes both “words” and “deeds” using Z-scores within the same industry-year sample. Finally, by calculating the difference between the two, this

paper constructs a continuous index of strategic decoupling in environmental disclosure (ED). The calculation formula is as follows:

$$ED_{i,t} = Z_Talk_{i,t} - Z_Action_{i,t} \quad (1)$$

When $ED_{i,t} > 0$, a larger value indicates that the firm is “overstating” relative to its industry peers; its substantive actions fail to support its high-profile rhetoric, reflecting a strong greenwashing propensity. Conversely, when $ED_{i,t} < 0$ a smaller value (i.e., a larger negative magnitude) indicates that the firm is “understating” within its industry; it deliberately maintains a low profile despite undertaking substantial environmental investments, reflecting a strong brownwashing propensity.

3.1.2. Explanatory Variable

Performance Feedback. This study defines performance feedback (PF) as the discrepancy between a firm’s actual performance level and its organizational aspiration level, reflecting the degree of performance deviation when a firm fails to achieve its expected targets. Specifically, drawing on the research of Neckebrouck et al. (2025) and Goyal L (2025), return on assets (ROA) is used to represent actual corporate performance (P), and a third-order autoregressive model (AR3) is employed to estimate the firm’s performance aspirations, which serves as the basis for calculating performance feedback [20,21]. The specific calculation steps are as follows:

$$P_{i,t} = \alpha + \sum_{e=1}^3 \beta_e P_{t-e,i} + \varepsilon_{i,t} \quad (2)$$

The residual $\alpha_{i,t}$ represents the difference between firm i ’s expected performance and actual performance in period t . It is used as the proxy variable for historical performance feedback and denoted as $PF_{i,t}$. When $PF_{i,t} > 0$, it indicates that the firm’s actual performance in the current period exceeds its expected performance; that is, when the firm’s operating conditions are better than its historical levels, it manifests as positive performance feedback (PPF). Conversely, when $PF_{i,t} < 0$, it manifests as negative performance feedback (NPF).

3.1.3. Moderator Variables

(1) Government Regulation

As an official document with the nature of a policy platform, the government work report systematically articulates the government’s priorities within a specific period. Drawing on the approach of Lin & Zhang (2023), this study measures the intensity of government environmental regulation by taking the natural logarithm of the word count of sentences containing environment-related keywords in the government work reports of various prefecture-level cities plus one [22]. This metric is then matched with the registered locations of listed companies to gauge the intensity of government regulation faced by the firms.

(2) Media Supervision

Existing literature predominantly measures media attention by obtaining the frequency of news reports related to a firm through internet news search engines. Drawing on the research of He et al. (2024), this paper utilizes the sum of the number of media report headlines from both quantitative statistics of financial news in newspapers and quantitative statistics of online news, sourced from the Chinese Research Data Services Platform (CNRDS), plus one, and takes the natural logarithm to measure the intensity of media attention received by heavily polluting enterprises [23].

3.1.4. Control Variables

The control variables selected in this paper include firm size, firm age, current asset ratio, number of employees, revenue growth rate, capital intensity, ownership check-and-balance, and CEO duality. Detailed definitions of the variables are presented in Table 1.

Table 1. Variable Definitions.

Variable Name	Variable Symbol	Variable Definition
Environmental Decoupling	ED	The detailed calculation is shown in the formula and description of the dependent variable ED in this section.
Greenwashing	GW	When ED > 0;
Brownwashing	GB	When ED < 0
Performance Feedback	PF	The detailed calculation is shown in the formula and description of the independent variable PF in this section.
Negative Performance Feedback	NPF	When PF < 0;
Positive Performance Feedback	PPF	When PF > 0
Media Coverage	Media	Natural logarithm of one plus the number of online and media report headlines.
Government Regulation	Gov	Natural logarithm of one plus the number of words in sentences containing environmental protection keywords in government work reports.
Firm Size	Size	Natural logarithm of total assets.
Firm Age	Age	Natural logarithm of one plus the current year minus the firm's establishment year.
Current Ratio	LIQ	Ratio of current assets to total assets.
Employees	EMP	Natural logarithm of the number of employees.
Revenue Growth Rate	Growth	(Current period operating revenue—Corresponding prior year period operating revenue)/Corresponding prior year period operating revenue.
Capital Intensity	CI	Ratio of total assets to total operating income.
Ownership Balance	Equity	Shareholding ratio of the 2nd to 5th largest shareholders divided by that of the largest shareholder.
CEO Duality	Dual	A dummy variable equal to 1 if the chairman and CEO are the same person, and 0 otherwise.

3.2. Model Design

This paper employs an ordinary least squares (OLS) panel data model controlling for multiple fixed effects for baseline parameter estimation, and constructs the following model (1) to examine the impact of performance feedback on corporate strategic decoupling in environmental disclosure. Drawing on numerous studies concerning performance feedback, this paper lags the explanatory and control variables by one period.

$$ED_{i,t} = \alpha_0 + \alpha_1 PF_{i,t-1} + \sum \text{Control}_{i,t-1} + \theta_k + \sigma_t + \varepsilon_{i,t} \quad (3)$$

where $ED_{i,t}$ represents the environmental decoupling, and $PF_{i,t-1}$ encompasses negative performance feedback (NPF) and positive performance feedback (PPF). The subscripts i and t denote firm and year, respectively. θ_k and σ_t represent industry and year fixed effects, respectively. $\varepsilon_{i,t}$ is the random disturbance term. In addition, all regressions in this paper employ robust standard errors clustered at the industry level, and $\text{Control}_{i,t}$ represents a series of control variables, the detailed definitions of which are provided in

Table 1. Furthermore, to test the moderating effects of government regulation and media attention, models (2) and (3) are specified:

$$ED_{i,t} = \alpha_0 + \alpha_1 PF_{i,t-1} + \alpha_2 M_{i,t-1} \times PF_{i,t-1} + \alpha_3 M_{i,t-1} + \beta_k \sum \text{Control}_{i,t-1} + \theta_k + \sigma_t + \varepsilon_{i,t} \quad (4)$$

In Equation (4) M includes government regulation and media attention.

3.3. Data Sources and Descriptive Statistics

This study selects A-share companies listed on the Shanghai and Shenzhen Stock Exchanges from 2011 to 2024. Listed companies with severe missing financial data are excluded, and data processing procedures such as winsorization and the removal of outliers are applied to the sample data, ultimately yielding 28,203 firm-year observations from 3673 enterprises. The descriptive statistics for all variables are presented in Table 2. The standard deviation of environmental decoupling is 1.249, with a minimum value of -4.626 and a maximum value of 3.797 . This indicates significant heterogeneity in strategic decoupling behaviors in environmental disclosure among different listed companies, which is perfectly aligned with the research context of this paper. The mean of performance feedback for the full sample is 0, conforming to the statistical properties of the residuals from the third-order autoregressive model employed in this study. This demonstrates that the expected performance measure constructed herein is unbiased, capable of accurately capturing the endogenous performance aspirations of corporate decision-makers, and that the measurement method is methodologically sound. The standard deviation of performance feedback is 0.060, with a minimum value of -0.412 and a maximum value of 0.424 , indicating substantial variation in the discrepancies between actual and expected performance among different listed companies, thereby providing sufficient variance for testing the core main effects in this paper. Overall, the values of all variables in this study fall within a reasonable range, free from extreme outliers. The sample distribution highly matches the overall operational and governance characteristics of A-share listed companies, laying a reliable data foundation for subsequent empirical analyses, including baseline regressions and moderation effect tests.

Table 2. Descriptive Statistics.

Variable Name	Sample Size	Mean	Standard Deviation	Minimum	Maximum
ED	28,203	0.004	1.249	-4.626	3.797
Greenwashing	12,243	0.948	0.924	0.000	3.797
Brownwashing	15,960	-0.719	0.946	-4.626	-0.000
Performance Feedback	28,203	0.000	0.060	-0.412	0.424
Negative Performance Feedback	11,852	-0.043	0.059	-0.412	-0.000
Positive Performance Feedback	16,351	0.031	0.037	0.000	0.424
Media Coverage	28,203	4.211	1.339	0.000	6.644
Government Regulation	28,203	3.843	0.351	2.639	4.635
Firm Size	28,203	22.273	1.345	19.581	26.413
Firm Age	28,203	2.075	0.955	0.000	3.401
Current Ratio	28,203	0.572	0.204	0.092	0.956
Employees	28,203	7.603	1.305	4.304	11.233
Revenue Growth Rate	28,203	0.377	1.078	-0.786	7.991
Capital Intensity	28,203	2.805	2.673	0.395	18.416
Ownership Balance	28,203	0.761	0.614	0.029	2.766
CEO Duality	28,203	0.300	0.458	0.000	1.000

4. Empirical Results

4.1. Baseline Regression

Table 3 reports the baseline regression results. All models strictly control for year and industry fixed effects to absorb unobservable confounding factors from macroeconomic cycles and industry characteristics. Columns (1) and (2) of Table 3 first report the overall impact of performance feedback on strategic decoupling in environmental disclosure. Without and with the inclusion of control variables, the coefficients of PF are -0.5145 and -0.4035 , respectively, both significantly negative at the 1% level. This indicates that as a firm's overall performance improves, its tendency of environmental decoupling of words and deeds shifts toward brownwashing. Conversely, as its overall performance deteriorates, its tendency of environmental decoupling of words and deeds shifts toward greenwashing. The behavioral theory of the firm emphasizes that corporate strategic logic exhibits asymmetry between favorable and adverse circumstances; conflating positive and negative feedback may obscure the true motivations of firms under different survival conditions. Therefore, this paper further splits the sample into a negative performance feedback group and a positive performance feedback group for a precise examination. Column (3) of Table 3 reports the impact of negative performance feedback on strategic decoupling in environmental disclosure. The results show that the regression coefficient for negative performance feedback is -1.2087 and significant at the 1% level. In the variable specification of this paper, negative performance feedback (NPF) is a negative value; a smaller value represents a larger shortfall from aspirations. Therefore, a larger negative performance feedback increases the likelihood of corporate greenwashing, supporting hypothesis H1a. Column (4) reports the impact of positive performance feedback (PPF) on strategic decoupling in environmental disclosure. The results show that the regression coefficient for PPF is -1.3052 , also significant at the 1% level. Since positive performance feedback is a positive value, a larger value represents greater profitability exceeding aspirations. Thus, this negative coefficient indicates that as positive performance feedback increases, the likelihood of corporate brownwashing increases, supporting hypothesis H1b.

Table 3. Baseline Regression Results.

	(1)	(2)	(3)	(4)
Variables	ED	ED	ED	ED
PF	-0.5145^{***} (0.1643)	-0.4035^{***} (0.1212)		
NPF			-1.2087^{***} (0.2479)	
PPF				-1.3052^{***} (0.2754)
Size		-0.3417^{***} (0.0331)	-0.3195^{***} (0.0443)	-0.3497^{***} (0.0326)
Age		0.1005^{***} (0.0339)	0.1000^{***} (0.0355)	0.1007^{**} (0.0395)
LIQ		0.3407 (0.3450)	0.0401 (0.5079)	0.5321 (0.4271)
EMP		-0.0935^{***} (0.0237)	-0.0970^{***} (0.0289)	-0.0985^{***} (0.0276)

Table 3. Cont.

	(1)	(2)	(3)	(4)
Variables	ED	ED	ED	ED
Growth		−0.0098 *** (0.0019)	−0.0071 ** (0.0028)	−0.0109 *** (0.0033)
CI		0.0018 (0.0088)	0.0163 (0.0098)	−0.0163 (0.0105)
Equity		−0.0386 *** (0.0106)	−0.0352 ** (0.0155)	−0.0473 *** (0.0143)
Dual		−0.0404 (0.0398)	−0.0402 (0.0445)	−0.0427 (0.0492)
Constant	6.4321 *** (1.5621)	8.2138 *** (0.7451)	7.7052 *** (0.8823)	8.4817 *** (0.7716)
Industry FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
N	28,203	28,203	11,852	16,351
R ² _a	0.3432	0.3964	0.3643	0.3902

Notes: Robust standard errors are in parentheses. **, and *** indicate statistical significance at the 5%, and 1% levels, respectively.

4.2. Robustness Checks

4.2.1. Replacing the Dependent Variable

To verify whether the core conclusions drawn from the environmental dimension possess broader generalizability, and to avoid measurement bias caused by the subjectivity of selecting a single dependent variable, this paper further expands the research perspective from the environmental (E) dimension to full-dimensional ESG washing behaviors for robustness checks. Drawing on the approaches of Zhang (2022) [24] and Hu et al. (2023) [11], this paper employs the difference between a firm's ESG information disclosure score and its actual ESG performance score as the proxy for corporate “washing” behavior. As shown in model (5), this study utilizes Bloomberg ESG disclosure scores ($ESG_{disi,t}$) and Sino-Securities Index ESG performance ratings ($ESG_{peri,t}$) as the primary data sources. First, the alphabetical Sino-Securities Index ESG ratings are converted into a 10-point numerical scale, with 10 being the highest. Subsequently, the Bloomberg disclosure scores and the converted Sino-Securities Index ESG scores are divided by 100 and 10, respectively, standardizing both into ratio indicators with a maximum value of 1. Next, these two ratio indicators are independently standardized by industry—that is, subtracting the industry mean and dividing by the industry standard deviation—yielding standardized $ESG_{peri,t}$ and $ESG_{peri,t}$. Ultimately, the ESG-Wash value is the difference between the two, reflecting the degree to which a firm's ESG disclosure deviates relative to the industry average, which constitutes a relative measure of washing.

$$ESG\text{-}Wash_{i,t} = \left(\frac{ESG_{disi,t} - \overline{ESG_{dis}}}{\sigma_{dis}} \right) - \left(\frac{ESG_{peri,t} - \overline{ESG_{peri}}}{\sigma_{peri}} \right) \quad (5)$$

When ESG-Wash is positive, it indicates that the firm's ESG information disclosure level exceeds its substantive performance, exhibiting an “overstating” greenwashing characteristic; a larger value signifies a higher degree of greenwashing. If ESG-Wash is negative,

it implies that the firm's disclosure level is lower than its actual performance, presenting an "understating" brownwashing characteristic; a smaller value indicates a higher degree of brownwashing. Due to missing data in both Bloomberg and Sino-Securities Index datasets, the valid sample size for ESG-Wash is 10,221. Columns (1) and (2) of Table 4 present the regression results. As the results show, both positive and negative performance feedback are inversely proportional to the degree of corporate ESG washing; that is, positive performance feedback is more likely to lead to brownwashing, while negative performance feedback is more inclined toward greenwashing, which is consistent with the previous conclusions.

Table 4. Regression Results of Replacing the Dependent Variable.

	(1)	(2)	(3)	(4)	(5)
Variables	ESG-Wash	ESG-Wash	ESG-Wash	ED_1	ED_1
PF	−1.2453 *** (0.3655)				
NPF		−1.1688 *** (0.4483)		−1.4242 *** (0.2747)	
PPF			−0.4649 ** (0.2182)		−2.0395 *** (0.3652)
Constant	−0.5156 *** (0.1222)	−0.4998 *** (0.1183)	−0.4872 *** (0.1232)	6.1832 *** (1.9112)	6.1327 *** (1.7104)
Control FE	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES
N	10,221	5558	4663	11,852	16,351
R ² _a	0.2383	0.3445	0.2277	0.3432	0.4212

Note: ** $p < 0.01$, *** $p < 0.001$.

Existing research indicates that there is often a time lag in translating a firm's substantive environmental actions into effective and substantive information disclosure. To rigorously rule out the potential interference of this time-lag effect on the core conclusions of this paper, drawing on the practices of relevant literature, this study lags the proxy variable for "deeds" by one period, denoted as ED_1. Table 4, Columns (4) and (5), report the regression results using the lagged action variable. Column (4) shows that the coefficient of negative performance feedback is −1.4242, significantly negative at the 1% level, indicating that even when accounting for the lag in actions, negative performance feedback still significantly promotes firms' greenwashing propensity of "talking much but doing little." Column (5) shows that the coefficient of positive performance feedback is −2.0395, also significantly negative at the 1% level, indicating that positive performance feedback still significantly reinforces firms' brownwashing behaviors of "talking little but doing much." The above results demonstrate that, after fully considering the time lag in translating environmental actions into information disclosure, the core conclusion of this paper—that performance feedback drives corporate two-way environmental decoupling of words and deeds—remains highly robust.

4.2.2. Replacing the Independent Variable

To avoid the potential interference of subjectivity in measuring the core explanatory variable on the research conclusions, this paper conducts a robustness check by altering the measurement method of performance feedback. In the baseline regression, this study employed the residual values from a third-order autoregressive model to measure perfor-

mance feedback, which primarily reflects a firm's historical comparison based on its own trajectory. However, a firm's behavioral decisions are influenced not only by its historical performance but also profoundly driven by its relative competitive position within the industry. Therefore, this paper employs an alternative measure. Specifically, return on assets (ROA) is used to represent actual corporate performance, and expected performance (A) is constructed through a linear combination of historical expected performance (HA) and social expected performance (SA). The specific form is shown in Equations (6) and (7):

$$A_{i,t} = \alpha_1 HA_{i,t} + (1 - \alpha_1) SA_{i,t} \quad (6)$$

$$PF_{z_{i,t}} = ROA_{i,t} - A_{i,t} \quad (7)$$

where historical expected performance (HA) is measured by the firm's return on assets in the previous year; social expected performance (SA) takes the arithmetic mean of the ROA of other firms in the same industry and year (excluding the focal firm). The range for the weight α_1 is set from 0 to 1, traversed with a step size of 0.1, selecting the α_1 that yields the maximum "Log-likelihood" value. In the results reported in this paper, α_1 is set to 0.5. The direction of performance feedback can be determined by the sign of $PF_{z_{i,t}}$, if $PF_{z_{i,t}}$ is less than 0, it indicates that the firm's actual performance has not reached the expected level, constituting negative performance feedback; conversely, it is positive performance feedback. The results are shown in columns (1) and (2) of Table 5, which are consistent with the previous findings.

Table 5. Regression Results of Replacing the Independent Variable.

	(1)	(2)	(3)	(4)
Variables	ED	ED	ED	ED
PF_z	−0.5861 *** (0.0839)	−0.5173 *** (0.1190)		
ROA_gap			−0.4908 ** (0.2459)	−1.0384 *** (0.1390)
Constant	−0.0306 *** (0.0114)	−0.0372 *** (0.0105)	−0.0181 ** (0.0570)	−0.0172 *** (0.0045)
Control FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
N	12,539	15,664	13,682	14,521
R ² _a	0.2417	0.2815	0.3172	0.3218

Note: ** $p < 0.01$, *** $p < 0.001$.

A firm's behavioral decisions are influenced not only by its historical performance but also profoundly driven by its relative competitive position within the industry. In light of this, based on the perspective of social comparison, this paper constructs an industry expected performance feedback indicator (ROA_gap) using the difference between the industry average return on assets and the firm's ROA. The results are shown in columns (3) and (4) of Table 5, which are consistent with the previous findings. Therefore, the research conclusions of this paper are not substantively affected by the measurement methods of the core variables, demonstrating good robustness.

4.2.3. Ruling out Alternative Hypotheses

The core logic of this paper is that performance aspiration discrepancies promote corporate strategic decoupling in environmental disclosure. However, in traditional corporate social

responsibility and ESG research, the dominant theory is the slack resources hypothesis. This hypothesis posits that ESG is a luxury good. The more profitable a firm is, the more slack resources it has to invest in ESG; conversely, the more a firm loses money, the less it can afford to engage in ESG. If this alternative hypothesis holds, then a firm's financial performance (ROA) and ESG performance should exhibit a strictly linear positive correlation. To rule out this confounding factor, this paper replaces the core explanatory variable with the firm's absolute return on assets (ROA) in the current period and its squared term (ROA^2), and replaces the dependent variable with ESG performance for regression testing. To avoid reverse causality, this paper lags ROA and its squared term by one period. The control variables are consistent with those in the baseline model.

Table 6 reports the regression results excluding the alternative hypothesis. Column (1) shows that the linear coefficient of ROA is significantly positive at the 1% level, indicating that, without considering the non-linear relationship, an increase in absolute performance is generally accompanied by an increase in washing propensity, which supports the slack resources hypothesis. However, in Column (2), after adding the squared term of ROA, the linear term of ROA remains significantly positive at 0.2886, but the coefficient of the squared term of ROA is significantly negative at the 1% level (-0.3663). This result indicates that there is indeed a significant inverted U-shaped relationship between a firm's ROA and its ESG performance. On the left side of the inverted U-shape, the firm is operating at a loss (corresponding to negative performance feedback). At this point, the firm faces a survival crisis and fundamentally lacks the financial resources to engage in substantive environmental protection; it can only resort to greenwashing through "much talk and little action," which corroborates H1a. On the right side of the inverted U-shape, the firm is profitable (positive performance feedback). Logically, they have the money to invest in ESG, but why does their public ESG performance decline instead? Because, to evade green costs and supervisory pressure, they opt for brownwashing, which perfectly corroborates H1b. In summary, although the slack resources hypothesis can partially explain corporate ESG behaviors, the behavioral theory of the firm provides a more precise and profound explanation of the psychological anchors and decision-making logic of management oscillating between greenwashing and brownwashing. The core mechanism of this paper is not simply resource-driven; the alternative hypothesis is effectively ruled out, and the core conclusions remain robust.

Table 6. Regression Results of Ruling Out Alternative Hypotheses.

	(1)	(2)
Variables	ESG	ESG
ROA	0.2979 ***	0.2886 ***
	(0.0716)	(0.0637)
ROA^2		-0.3663 ***
		(0.0760)
Constant	4.2739 ***	4.2765 ***
	(1.1324)	(1.1018)
Control FE	YES	YES
Industry FE	YES	YES
Time FE	YES	YES
N	28,203	28,203
R^2_a	0.2253	0.3282

Note: *** $p < 0.001$.

4.2.4. Construct Validity of the Dependent Variable

This paper constructs a continuous index of strategic decoupling in environmental disclosure by separately measuring firms' "words" and "deeds" and calculating their difference. However, in empirical measurement, the word frequency in corporate annual reports may be influenced by factors such as industry-specific jargon, a firm's inherent reporting style, or an overall low willingness to disclose, implying the existence of measurement noise. On the one hand, a negative brownwashing value ($ED < 0$) might simply be due to a firm's overall low willingness to disclose environmental issues. On the other hand, a positive greenwashing value ($ED > 0$) might merely result from a firm's habit of lengthy reporting or being driven by universal industry environmental templates. To thoroughly rule out the interference of the aforementioned mechanical artifacts and establish the construct validity of strategic decoupling in environmental disclosure, this paper introduces external evidence independent of the baseline model (green patents)—the environmental (E) dimension performance score evaluated by an authoritative third-party agency (Sino-Securities)—to objectively measure firms' substantive environmental efforts. Combining this with the absolute environmental disclosure word frequency (Talk), this paper conducts a grouped mean difference test (T-test) between the brownwashing group and the greenwashing group. Table 7 reports the test results for construct validity. The data show that firms classified by this paper into the "brownwashing group" have an independent mean substantive environmental score (61.526) that is significantly higher than that of the greenwashing group at the 1% level, with the mean difference being significantly positive ($T = 3.49$). Simultaneously, the mean absolute environmental disclosure word frequency (0.010) of the brownwashing group in annual reports is significantly lower than that of the greenwashing group (0.059) at the 1% level, exhibiting a cliff-like drop. The ED index constructed in this paper can extremely precisely capture corporate greenwashing or brownwashing behaviors on environmental issues in the capital market.

Table 7. Grouped Mean T-test.

	(1)	(2)	(3)
Variables	BW Group ($ED < 0$)	GW Group ($ED \geq 0$)	Mean Difference (T-Value)
Independent Environmental Performance	61.526	59.196	2.330 *** (3.49)
Absolute Environmental Disclosure	0.010	0.059	−0.049 *** (−24.53)

Note: *** $p < 0.001$.

4.2.5. Adding Firm Fixed Effects

In the baseline regression, this paper controls for industry and year fixed effects to absorb the confounding influences of macroeconomic cycle shocks and universal industry disclosure templates. However, a firm's environmental information disclosure strategies and green innovation decisions may also be influenced by time-invariant, unobservable individual characteristics. To mitigate the endogeneity bias caused by such omitted variables, this paper further introduces firm individual fixed effects into the baseline model. Table 8 reports the regression results after controlling for firm individual fixed effects. Columns (1) to (3) show that the coefficients of overall performance feedback, NPF and PPF are significantly negative at the 1% level. This indicates that, even after eliminating the interference of time-invariant firm heterogeneity, the core conclusion that performance feedback drives corporate strategic decoupling in environmental disclosure remains highly robust.

Table 8. Adding Fixed Effects.

	(1)	(2)	(3)
Variables	ED	ED	ED
PF	−0.5383 *** (0.0839)		
NPF		−1.2173 *** (0.3190)	
PPF			−1.1908 *** (0.2459)
Constant	−3.5306 *** (1.1214)	−4.0372 *** (0.9765)	−3.8181 *** (1.2371)
Control FE	YES	YES	YES
Industry FE	YES	YES	YES
Time FE	YES	YES	YES
Firm FE	YES	YES	YES
N	28,203	11,852	16,351
R ² _a	0.3917	0.3815	0.3172

Note: *** $p < 0.001$.

4.3. Mechanism Tests

4.3.1. Government Regulation

Based on the preceding theoretical analysis, the transmission mechanism of the moderating effect of government regulation operates through two channels. On the one hand, in the context of negative performance feedback, government regulation relies on penetrating audits and substantive penalties to sharply elevate the costs of misrepresentation and reduce the benefits of greenwashing, thereby weakening the positive relationship between negative performance feedback and corporate greenwashing. On the other hand, in the context of positive performance feedback, stringent government regulation tightens the mandatory compliance boundaries of environmental information disclosure, which stimulates the defensive motives of highly profitable firms to evade “whipping the fast ox,” thereby strengthening the positive relationship between positive performance feedback and corporate brownwashing.

Table 9 reports the regression results for the moderating effect of government regulation. Column (1) presents the overall moderating effect of government regulation using the full sample. The coefficient for performance feedback is −1.4391, significantly negative at the 1% level, which is consistent with the core findings of the baseline regression. The coefficient for the interaction term between performance feedback and government regulation is 0.2407, significantly positive at the 1% level. This indicates that government regulation significantly attenuates the positive impact of negative performance feedback on corporate greenwashing and amplifies the positive impact of positive performance feedback on corporate brownwashing, providing preliminary validation for the moderating effect of government regulation. Column (2) examines the moderating role of government regulation under negative performance feedback. The coefficient for negative performance feedback is −0.8457, significant at the 1% level, and the coefficient for the interaction term is significantly positive at the 1% level. This demonstrates that as the intensity of regional government regulation increases, the positive effect of negative performance feedback on corporate greenwashing propensity is significantly suppressed, lending empirical support

to hypothesis H2a. Column (3) examines the moderating role of government regulation under positive performance feedback. The results show that the coefficient for positive performance feedback is -0.7168 , significant at the 1% level, and the coefficient for the interaction term is significantly negative at the 1% level. This indicates that government regulation significantly magnifies the positive effect of positive performance feedback on corporate brownwashing, providing empirical support for hypothesis H2b.

Table 9. Regression Results of Government Regulation.

	(1)	(2)	(3)
Variables	ED	ED	ED
PF	-1.4391^{***} (0.4282)		
PF $\times$ Gov	0.2407^{***} (0.0489)		
Gov	-0.2588^{***} (0.0788)	-0.2159^{***} (0.0707)	-0.2033^{***} (0.0753)
NPF		-0.8457^{***} (0.2508)	
NPF $\times$ Gov		0.6429^{***} (0.1691)	
PPF			-0.7168^{***} (0.2558)
PPF $\times$ Gov			-0.6789^{***} (0.2592)
Constant	1.1667^{***} (0.1888)	1.7584^{***} (0.2705)	1.1441^{***} (0.2935)
Control FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
N	28,203	11,852	16,351
R ² _a	0.3425	0.4232	0.3864

Note: *** $p < 0.001$.

To further characterize coercive legitimacy pressure from the perspective of substantive enforcement deterrence, this paper employs the natural logarithm of the number of environmental penalty cases at the prefecture-level city plus one as a proxy variable for the enforcement intensity of government regulation (Penal) in an auxiliary test. The data is sourced from the total number of annual environmental penalty cases across prefecture-level cities nationwide published by the Peking University Chinalawinfo judicial case retrieval system. A higher number of environmental penalty cases in a region signifies stronger local government crackdowns on environmental violations, resulting in higher costs for corporate environmental information manipulation and non-compliance. Table 10 reports the regression results using environmental penalties as the moderating variable. Column (1) shows that, in the full sample, the coefficient of the interaction term between overall performance feedback and environmental penalties is significantly positive at the 1% level. Column (2) shows that the coefficient of the interaction term between

negative performance feedback and environmental penalties is significantly positive at the 1% level; its sign is opposite to that of the main effect, supporting hypothesis H2a. Column (3) shows that the coefficient of the interaction term between positive performance feedback and environmental penalties is significantly negative at the 5% level. Its sign is consistent with that of the main effect, supporting hypothesis H2b.

Table 10. Regression Results of Intensity of Environmental Penalties.

	(1)	(2)	(3)
Variables	ED	ED	ED
PF	−0.7947 *** (0.3014)		
PF × Penal	0.0947 *** (0.0294)		
Penal	−0.0198 *** (0.0063)	−0.0229 *** (0.0072)	−0.0214 *** (0.0057)
NPF		−1.6441 *** (0.5184)	
NPF × Penal		0.0647 *** (0.0211)	
PPF			−1.9228 ** (0.7931)
PPF × Penal			−0.1435 ** (0.0583)
Constant	2.3344 *** (0.6415)	3.1354 *** (0.5613)	3.0872 *** (0.7523)
Control FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
N	28,203	11,852	16,351
R ² _a	0.3162	0.4058	0.4194

Note: ** $p < 0.01$, *** $p < 0.001$.

4.3.2. Media Attention

Based on the preceding theoretical analysis, the transmission mechanism of the moderating effect of media attention operates through two channels. On the one hand, for underperforming firms mired in negative performance feedback, media attention, leveraging traffic leverage and agenda-setting, vastly elevates the benefits of corporate green falsification. It provides firms with a stage to cater to public sentiment, thereby strengthening the positive relationship between negative performance feedback and corporate greenwashing. On the other hand, for firms experiencing positive performance feedback, high media attention compresses the space for concealing environmental information and amplifies the reputational benefits of substantive green innovation and environmental actions, thereby weakening the positive relationship between positive performance feedback and corporate brownwashing.

Table 11 reports the regression results for the moderating effect of media attention. Column (1) presents the overall moderating effect of media attention using the full sample.

The coefficient for performance feedback is -0.5234 , significantly negative at the 1% level, which is consistent with the core findings of the baseline regression. The coefficient for the interaction term between performance feedback and media attention is -0.2341 , significantly negative at the 1% level. This indicates that media attention significantly strengthens the positive impact of negative performance feedback on corporate greenwashing and attenuates the positive impact of positive performance feedback on corporate brownwashing, providing preliminary validation for the moderating effect of media attention. Column (2) examines the moderating role of media attention under negative performance feedback. The coefficient for negative performance feedback is -0.7623 , significant at the 1% level, and the coefficient for the interaction term is significantly negative at the 1% level. This demonstrates that as the intensity of media attention increases, the positive effect of negative performance feedback on corporate greenwashing propensity is significantly magnified, lending empirical support to hypothesis H3a. Column (3) examines the moderating role of media attention under positive performance feedback. The results show that the coefficient for positive performance feedback is -0.7352 , significant at the 1% level, and the coefficient for the interaction term is significantly positive at the 5% level. This indicates that media attention significantly weakens the positive effect of positive performance feedback on corporate brownwashing, providing empirical support for hypothesis H3b.

Table 11. Regression Results of Media Attention.

	(1)	(2)	(3)
Variables	ED	ED	ED
PF	-0.5234^{***} (0.1567)		
PF $\times$ Media	-0.2341^{***} (0.0799)		
Media	-0.1567^{***} (0.0171)	-0.1659^{***} (0.0301)	-0.1738^{***} (0.0192)
NPF		-0.7623^{***} (0.2503)	
NPF $\times$ Media		-0.3841^{***} (0.0814)	
PPF			-0.7352^{***} (0.2301)
PPF $\times$ Media			0.3054^{**} (0.1207)
Constant	0.5912^{***} (0.0713)	0.6004^{***} (0.1213)	0.5631^{***} (0.0804)
Control FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
N	28,203	11,852	16,351
R ² _a	0.3126	0.3242	0.3865

Note: ** $p < 0.01$, *** $p < 0.001$.

Synthesizing the above results of the moderation effect tests, this paper reveals the heterogeneous governance mechanisms of external legitimacy pressures in corporate two-way strategic decoupling in environmental disclosure. The research demonstrates that external supervisory forces do not serve as a homogeneous constraint mechanism deterring corporate opportunistic behaviors; rather, their governance efficacy is profoundly contingent upon how their power attributes asymmetrically intervene in firms' cost-benefit expectations regarding environmental information manipulation. Specifically, coercive legitimacy pressure, represented by government regulation, essentially acts as a magnifier of the costs of misrepresentation in corporate environmental decision-making. Its potent punitive deterrence effectively suppresses the greenwashing attempts of underperforming firms, but simultaneously exacerbates the motives of high-performing firms to evade "political costs" and the "whipping the fast ox" effect, thereby inducing more deep-seated brownwashing behaviors. In contrast, normative legitimacy pressure, represented by media attention, essentially functions as a magnifier of the performative benefits in corporate green image management. Its high exposure dismantles the invisibility strategies of high-performing firms, compelling them to publicly disclose their substantive investments (thereby suppressing brownwashing). However, its traffic-driven nature, coupled with a lack of penetrating verification, provides underperforming firms with a stage to cater to public sentiment, thereby catalyzing even more severe greenwashing. This finding not only deconstructs the unidimensional assumption regarding the governance effects of legitimacy pressure in neo-institutional theory but also provides a novel theoretical perspective for understanding corporate environmental disclosure strategies within complex institutional environments.

4.4. Heterogeneity Analysis

4.4.1. Industry Pollution Intensity

Industry pollution intensity is one of the core determinants of corporate legitimacy pressure. As the primary targets of environmental regulation, heavily polluting industries face more stringent compliance requirements and higher stakeholder attention regarding their environmental disclosures and actions. Compared to enterprises in non-heavily polluting industries, there are fundamental differences in the cost-benefit expectations based on which these two types of firms adjust their environmental strategies in response to performance feedback. Therefore, referring to the Guidelines for the Industry Classification of Listed Companies issued by the CSRC and the Classified Management Directory of Environmental Protection Verification of Listed Companies issued by the Ministry of Ecology and Environment, this paper divides the sample into heavily polluting and non-heavily polluting industry groups for subsample regression tests. The results are presented in Table 12. Columns (1) and (2) report the regression results of negative performance feedback across industries with different pollution intensities. The results show that the coefficients for negative performance feedback in both non-heavily polluting and heavily polluting industries are significantly negative at the 1% level, and the absolute value of the coefficient for heavily polluting industries is significantly larger than that for non-heavily polluting industries. This result indicates that the positive impact of negative performance feedback on corporate greenwashing propensity is more pronounced in heavily polluting industries. Columns (3) and (4) report the regression results of positive performance feedback across industries with different pollution intensities. The results show that in the non-heavily polluting industry sample, the coefficient for positive performance feedback is significantly negative at the 1% level, consistent with the main effect in the baseline regression. This indicates that in non-heavily polluting industries, positive performance feedback significantly increases corporate brownwashing propensity. However, in the heavily polluting industry sample, the coefficient for positive performance feedback is significantly positive at the

5% level, indicating that in heavily polluting industries, positive performance feedback actually significantly suppresses corporate brownwashing propensity. The core reason for this heterogeneous result is that enterprises in heavily polluting industries are constantly under the spotlight of environmental regulation and public scrutiny. Even when their operating performance exceeds expectations, they still need to maintain their legitimacy status and evade regulatory inquiries and public skepticism through adequate environmental information disclosure. Consequently, they will proactively elevate their environmental disclosure levels to align with regulatory requirements.

Table 12. Regression Results of Industry Pollution Intensity.

	(1)	(2)	(3)	(4)
Variables	Non-Heavy	Heavy	Non-Heavy	Heavy
NPF	−1.5456 *** (0.3469)	−1.8203 *** (0.5640)		
PPF			−2.5491 *** (0.5239)	1.6254 ** (0.6039)
Constant	−0.0811 *** (0.0139)	−0.0786 *** (0.0186)	−0.1806 *** (0.0150)	−0.0885 *** (0.0181)
Control FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
N	8237	3615	12,097	4254
R ² _a	0.3324	0.3134	0.3123	0.3505

Note: ** $p < 0.01$, *** $p < 0.001$.

4.4.2. Ownership Structure

The nature of property rights is a core institutional factor that determines corporate operational goals, legitimacy foundations, and the logic of strategic choices. State-owned enterprises (SOEs) pursue both profitability and policy objectives; they inherently possess a stronger legitimacy foundation and face relatively weaker market reputation constraints and profitability pressures. In contrast, the operation and development of Private-owned enterprises (POE) rely heavily on market recognition and profitability. Performance fluctuations exert a much more significant impact on their legitimacy status, financing capabilities, and market reputation. Consequently, there are fundamental differences in the cost–benefit expectations based on which these two types of firms adjust their ESG strategies in response to performance feedback. Therefore, this paper divides the sample based on the nature of the controlling shareholder: if the controlling shareholder includes state-owned entities, the firm is defined as an SOE; the remaining sample is defined as a POE. The subsample regression results are presented in Table 13. The results in Columns (1) and (2) show that in the POE sample, the coefficient for negative performance feedback is significantly negative at the 1% level, consistent with the main effect in the baseline regression. However, in the SOE sample, the coefficient for negative performance feedback is −0.5731 and statistically insignificant, indicating that negative performance feedback has no significant impact on the greenwashing propensity of SOEs. This suggests that SOE executives face more stringent political discipline audits, making their costs of misrepresentation far higher than those of private enterprises. Therefore, SOEs lack the motivation to engage in high-risk greenwashing behaviors for legitimacy hedging during adverse circumstances.

Table 13. Regression Results of Ownership Structure.

	(1)	(2)	(3)	(4)
Variables	POE	SOE	POE	SOE
NPF	−1.0527 *** (0.2844)	−0.5731 (0.6957)		
PPF			−0.5275 *** (0.1819)	1.2523 *** (0.2683)
Constant	−0.0466 *** (0.0124)	−0.1690 *** (0.0291)	−0.0567 *** (0.0143)	−0.0248 *** (0.0075)
Control FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
N	6830	5022	9351	7000
R ² _a	0.3218	0.3053	0.2985	0.3524

Note: *** $p < 0.001$.

The results in Columns (3) and (4) show that in the POE sample, the coefficient for positive performance feedback is -0.5275 , significantly negative at the 1% level. Conversely, in the SOE sample, the coefficient for positive performance feedback is 1.2523 , significantly positive at the 1% level. This indicates that positive performance feedback actually significantly suppresses the brownwashing propensity of SOEs and drives them to elevate their environmental disclosure levels. When SOEs are in favorable circumstances, extensively publicizing their substantive contributions in environmental protection and social responsibility is not only an inevitable requirement for fulfilling their political missions but also a core pathway for SOE executives to accumulate political capital for promotion. Therefore, SOEs will absolutely not choose brownwashing in favorable circumstances; instead, they will actively highlight their sustainability achievements. Synthesizing the results from Table 10, the heterogeneity test based on ownership structure not only rules out other alternative explanations but also provides compelling evidence that corporate strategic decoupling in environmental disclosure is, in essence, a rational response derived from firms' precise calculation of costs and benefits.

4.4.3. Competitive Pressure

Industry competitive pressure is a core market environment factor that determines corporate operational strategies, performance sensitivity, and demands for market reputation. Under varying degrees of competitive pressure, the survival risks and market feedback resulting from performance fluctuations differ fundamentally, which in turn alters the cost–benefit expectations based on which firms adjust their environmental information disclosure strategies in response to performance feedback. In highly competitive industries, a firm's market share, investor confidence, and profitability stability are more sensitive to performance fluctuations, making the survival pressure brought about by performance shortfalls more pronounced. Conversely, in low-competition industries, firms possess stronger market pricing power and profitability stability, and the impact of performance fluctuations on their operations is relatively limited. Consequently, the logic behind the environmental information disclosure strategy choices of these two types of firms differs significantly. This paper employs the industry Lerner Index to measure the industry competitive pressure faced by firms. The Lerner Index reflects a firm's market pricing power; a smaller index indicates more intense industry competition, whereas a larger index indicates

stronger market power and lower industry competitive pressure. The subsample regression results are presented in Table 14. Columns (1) and (2) report the regression results of negative performance feedback across industries with different competitive pressures. The results show that the absolute value of the coefficient for the high competitive pressure group is significantly larger than that for the low competitive pressure group. This result indicates that the greater the industry competitive pressure, the stronger the positive impact of negative performance feedback on corporate greenwashing propensity.

Table 14. Regression Results of Competitive Pressure.

	(1)	(2)	(3)	(4)
Variables	Low	Intense	Low	Intense
NPF	−1.4856 *** (0.4895)	−1.7209 *** (0.3686)		
PPF			−2.3719 *** (0.5014)	−2.2047 *** (0.5193)
Constant	−0.0774 *** (0.0190)	−0.0828 *** (0.0136)	−0.1318 *** (0.0150)	−0.1804 *** (0.0145)
Control FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
N	6800	5052	11,822	4529
R ² _a	0.3223	0.2908	0.3021	0.3175

Note: *** $p < 0.001$.

The results in Columns (3) and (4) show that the coefficient for positive performance feedback in the high competitive pressure group is −2.2047, significantly negative at the 1% level, and its absolute value is significantly smaller than that in the low competitive pressure group. This result indicates that the smaller the industry competitive pressure, the stronger the positive impact of positive performance feedback on corporate brownwashing propensity; conversely, high competitive pressure significantly weakens the driving effect of positive performance feedback on corporate brownwashing. The heterogeneity test of industry competitive pressure once again proves that the “decoupling of words and deeds” in corporate environmental information disclosure is a rational strategic choice made by firms under specific market competition landscapes to maximize their probability of survival and safeguard their vested interests.

5. Discussion

By constructing an index of environmental decoupling of words and deeds (ED), this paper systematically examines the asymmetric impact of performance feedback on corporate strategic decoupling in environmental disclosure (greenwashing and brownwashing) and reveals the intervention mechanisms of heterogeneous legitimacy pressures therein. The findings of this study engage in a profound dialogue with and expand upon recent relevant empirical literature.

Breaking Through the Linear Cognition of “Unidirectional Greenwashing” and the “Slack Resources Hypothesis”. Existing empirical research on corporate environmental information manipulation, mostly based on agency theory or the slack resources hypothesis, unidirectionally explores why firms “overstate” (greenwash) [1,5]. These studies often arrive at the linear conclusion that “the scarcer the financial resources, the stronger

the greenwashing motivation.” This paper’s finding that negative performance feedback promotes greenwashing provides the latest micro-level evidence from the Chinese capital market to support this classic assertion. However, the core breakthrough of this paper lies in challenging the traditional cognition that “the more abundant the resources, the more transparent the disclosure.” This study discovers that under positive performance feedback (favorable circumstances), highly profitable firms paradoxically generate a strong motivation to “talk little but do much” (brownwashing/green muting). This counter-intuitive finding significantly expands the theoretical framework of the “internal-external action gap” proposed by Hawn & Ioannou (2016) [25]. This paper points out that, in the context of the capital market, brownwashing is by no means harmless modesty; rather, it is an opportunistic behavior where high-performing firms, to evade the political apportionments of “whipping the fast ox” and excessively high public expectations, would rather endure a certain degree of “value penalty” in exchange for strategic security. This provides a novel asymmetric logic for understanding the complex environmental impression management of enterprises in emerging economies.

Deconstructing the “Myth of Homogeneity” in the Governance Effects of Legitimacy Pressure. When exploring external governance mechanisms, traditional neo-institutional literature largely assumes that legitimacy pressures (e.g., government regulation, media supervision) exert a unidirectional, positive governance effect on corporate compliance behaviors—that is, “the greater the pressure, the fewer the violations.” The findings of this paper deconstruct this “myth of homogeneity.” From the perspective of the actuarial “cost-benefit” analysis of environmental information manipulation, this paper, for the first time, distinguishes external pressures into “manipulation cost amplifiers” (government regulation) and “performative benefit amplifiers” (media attention). This paradigm shift elegantly explains why, in certain contexts, external monitoring not only fails to suppress environmental decoupling of words and deeds but may paradoxically induce more severe greenwashing performances (e.g., underperforming firms under high media attention) or exacerbate deeper strategic invisibility (e.g., high-performing firms under stringent government regulation). This aligns perfectly with recent literature emphasizing the “unintended consequences of regulation,” providing breakthrough theoretical insights for understanding government-business and media-business games in complex institutional environments.

6. Conclusions and Implications

6.1. Conclusions

Against the backdrop of the “Dual Carbon” goals and the advancement of Chinese modernization, preventing corporate environmental information manipulation and guiding capital toward social good are critical imperatives for the capital market. Using A-share listed companies in China as the research sample, this paper innovatively combines text analysis with objective data to construct a corporate strategic decoupling index in environmental disclosure. Drawing upon the behavioral theory of the firm and legitimacy theory, this study systematically investigates the driving mechanisms of performance feedback on corporate strategic decoupling in environmental disclosure, as well as the asymmetric intervention effects of heterogeneous legitimacy pressures therein. The findings are as follows:

First, performance aspiration discrepancies constitute the underlying driver of corporate strategic decoupling in environmental disclosure. Corporate greenwashing and brownwashing behaviors are not merely a moral issue but a strategic choice grounded in survival conditions. When facing negative performance feedback, driven by survival anxiety, firms tend to adopt a greenwashing strategy to fraudulently obtain legitimacy benefits. Conversely, when receiving positive performance feedback, to evade agency

costs and potential political costs, firms are inclined to adopt a brownwashing strategy for strategic invisibility.

Second, external legitimacy pressures exert asymmetric governance effects on corporate strategic decoupling in environmental disclosure. Government regulation, acting as a magnifier of misrepresentation costs, effectively suppresses the greenwashing attempts of underperforming firms through its potent deterrence; however, the high-pressure environment also exacerbates the fear of “whipping the fast ox” among high-performing firms, thereby amplifying their brownwashing motives. In contrast, media attention, acting as a magnifier of performative benefits, dismantles the invisibility strategies of high-performing firms and suppresses brownwashing through its high exposure; yet, its traffic-driven nature, lacking penetrating verification, provides underperforming firms with a stage to cater to public sentiment, thereby catalyzing even more severe greenwashing performances.

Third, the heterogeneity analysis further corroborates the micro-mechanisms of survival conditions and cost–benefit expectations. Greenwashing behaviors induced by negative feedback are primarily concentrated among private enterprises facing hard budget constraints and firms in highly competitive environments. Meanwhile, brownwashing behaviors induced by positive feedback are highly concentrated among private enterprises lacking a sense of political security and firms in low-competition (highly monopolistic) industries that easily attract unwanted attention (“tall trees catch the wind”).

6.2. Theoretical Contributions

The findings of this study make theoretical contributions in three primary areas. First, it expands the research boundaries of environmental information disclosure and strategic behaviors by constructing a unified analytical framework for two-way strategic decoupling in environmental disclosure. Most existing literature confines its focus to corporate greenwashing behaviors. This paper integrates both greenwashing and brownwashing—two types of strategic decoupling behaviors—into a unified analytical framework, clarifies the intrinsic logical differences between them, and reveals the fundamental nature of environmental information disclosure as a tool for corporate impression management, thereby enriching the application of impression management theory in the realm of non-financial information.

Second, it deepens the application context of the behavioral theory of the firm and reveals the asymmetric logic of information manipulation. Traditional research on the behavioral theory of the firm primarily focuses on the search for and allocation of tangible resources, such as R&D and M&A, paying less attention to firms’ strategic behaviors regarding non-financial information under performance fluctuations. This paper extends the analytical framework of the behavioral theory of the firm to the domain of non-financial information disclosure. It demonstrates that when firms face substantive environmental cost constraints, the strategic manipulation of environmental information disclosure becomes an alternative problemistic search solution to cope with performance shortfalls. Simultaneously, this paper delineates the asymmetric environmental information manipulation logic formed by firms based on different risk preferences in the domain of losses (negative performance feedback) and the domain of gains (positive performance feedback), thereby addressing the lack of a dynamic performance perspective in existing research on the antecedents of corporate green behaviors.

Finally, it analyzes the homogenizing myth regarding legitimacy pressure within neo-institutional theory. Existing research based on neo-institutional theory often implicitly assumes that legitimacy pressure exerts a unidirectional positive governance effect on corporate compliance behaviors. It generally treats external supervisory forces, such as gov-

ernment regulation and media attention, as homogeneous constraints that deter corporate opportunistic behaviors, insufficiently revealing the heterogeneous governance effects of different types of legitimacy pressure and their varying impacts across different corporate performance contexts. From the analytical perspective of the costs and benefits of misrepresentation, this paper clarifies the attribute differences between two core types of legitimacy pressure: coercive legitimacy pressure, represented by government regulation, essentially acts as a magnifier of misrepresentation costs for corporate environmental strategic behaviors; while normative legitimacy pressure, represented by media attention, essentially functions as a magnifier of performative benefits for corporate environmental image management. This provides novel theoretical insights for understanding the games between the government and enterprises, as well as between the media and enterprises, within complex institutional environments.

6.3. Practical Implications

The findings of this study offer important practical implications for improving the ESG ecosystem in China's capital market and mitigating the risks of strategic disclosure:

First, for regulatory authorities, it is essential to construct an ESG regulatory system that emphasizes both penetrating oversight and differentiated approaches. On the one hand, regulators must not only scrutinize the false prosperity of "underachievers" but also remain vigilant against the strategic invisibility of "top performers." They should accelerate the implementation of a mandatory ESG disclosure regime, unify quantitative disclosure standards, and compress the manipulation space for corporate ESG washing. On the other hand, when enforcing environmental regulations, local governments should be wary of the negative effects of "whipping the fast ox." Positive incentive mechanisms (e.g., tax exemptions, green channels) should be established to eliminate the additional political costs and apportionment risks that highly profitable firms might face for undertaking social responsibilities, thereby encouraging "doers" to speak up.

Second, for institutional and individual investors, it is crucial to beware of the dual traps of the media spotlight and short-term performance. When allocating ESG assets, investors should not blindly trust firms' rhetorically embellished PR reports or enthusiastic media coverage. Particularly when a firm has recently suffered a severe financial downturn yet aggressively touts its ESG vision in the media, this is highly likely a greenwashing performance designed to mask its financial collapse. Investors should pierce through the textual facade and anchor their focus on substantive "hard data," such as green patents and capitalized environmental expenditures.

Third, for corporate managers, they should abandon the speculative mindset of treating ESG as a short-term legitimacy hedging tool. Whether it is "overstating" in times of adversity or "understating" in times of prosperity, the decoupling of words and deeds will ultimately impair the firm's long-term reputational capital. Firms should deeply embed ESG principles into their core business models, achieving a fundamental shift from strategic coping to substantive value creation.

Fourth, it offers universal implications for global capital markets and environmental governance. Although the empirical tests in this paper are based on the Chinese context, the two-way decoupling logic of "brownwashing in favorable circumstances" and "greenwashing in adverse circumstances" possesses profound micro-foundations globally. On the one hand, "greenwashing in adverse circumstances," serving as a low-cost legitimacy hedging tool, is equally prevalent among European and American enterprises facing financial pressure. On the other hand, "brownwashing in favorable circumstances (green muting)" is by no means unique to China. In developed Western capital markets, with the rise in the "anti-ESG backlash" and the sharp increase in climate litigation risks, a growing number of

highly profitable multinational corporations have similarly opted for the strategic silence of “talking little but doing much” to guard against stringent “hypocrisy accusations (hypocrisy avoidance)” or to evade excessive scrutiny by radical environmentalists. Therefore, the asymmetric governance mechanism revealed in this paper—that “external monitoring (such as high media exposure) may dismantle invisibility or induce performative actions”—holds universal reference value for global regulatory bodies (such as the U.S. SEC and European EFRAG). It provides critical insights for formulating unified climate disclosure standards (e.g., avoiding the chilling effect triggered by mandatory disclosure) and preventing “environmental information arbitrage” by multinational corporations within global supply chains.

6.4. Research Limitations and Future Prospects

Although this paper has conducted rigorous theoretical deduction and empirical testing, certain limitations remain, warranting further deepening in future research:

First, there are textual dimension limitations regarding the measurement of “words” (disclosure). This paper primarily characterizes firms’ willingness to disclose by extracting the frequency of green keywords in corporate annual reports (MD&A). Although annual reports possess extremely high strategic authority, the channels for corporate impression management have become increasingly diversified in the digital age. Future research could introduce natural language processing (NLP) and machine learning technologies to further extract data from firms’ official social media (e.g., WeChat official accounts, Weibo), audio recordings of executive earnings briefings, and independent ESG reports. This would allow for the construction of a multimodal, omnichannel text analysis matrix to more comprehensively capture corporate washing dynamics.

Second, regarding the research context, there is a lack of comparison across cross-national institutional environments. The empirical sample of this study is confined to A-share listed companies in China. The phenomena discovered in this research—such as positive feedback inducing brownwashing and government regulation exacerbating brownwashing—are profoundly embedded within the specific context of government-business relations and political costs in China. Future research could extend this theoretical framework to cross-national samples, comparing how different institutional logics and cultural backgrounds reshape the antecedents and manifestations of corporate ESG washing in liberal market economies (e.g., the United States) versus coordinated market economies (e.g., Europe), thereby further testing the generalizability and boundary conditions of this theory.

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